Risk Management Assignment 2

Corporate Bonds & Single Factor Model

Politecnico di Milano — Financial Engineering
FY 2025/2026

01

Case Study

2 different bonds portfolios as of 15 February 2008

> *You will model default probability, intensity, recovery, and z-spread of some bonds issued by an IG company and you will model a portfolio made by several different bonds, all using the curve bootstrapped in Assignment 0.*

Portfolio Positions A

Two Defaultable Bonds positions held by Polimi Capital (valuation date: 15/02/2008)

1

Long Bond issued by Beta 1y

Maturity: 1 year **Coupon:** 5.5% semi-annual **Dirty price:** €100.00

Face value: €10 Mln

€10M

Notional

2

Long Bond issued by Beta 2y

Maturity: 2 years **Coupon:** 7% semi-annual **Dirty Price:** €103.00

Face value: €10 Mln

€10M

Notional

Conventions — Coupon semi-annual, 30/360 European, mod-foll, Beta Recovery=40%

Portfolio Positions B

100 Zero Coupon Bonds positions held by Polimi Capital (valuation date: 15/02/2008)

100 Bonds issued by 100 different IG companies

Maturity: 2 years **Coupon:** 0% **Price:** €tbd

Face Value: €1 Mln for each bond

€1M

Notional

Each company has recovery=40% and is affected by migration, default and correlation risk.

02

Key Concepts

Default, recovery, intensity Z-spread, transition matrix and single factor model

> *This section introduces the key building blocks you will work with. Refer to "Notes on Defaultable Bonds" for the mathematical details.*

Defaultable Bond

A bond issued by an entity that has default risk

What happens in case of default?

The issuer fails to meets its obligations

There are several definition of Default or Termination events (ISDA). The bondholder is entitled to demand immediate repayment of the principal amount.

Recovery

Part of the investment that the investor can recover

After the default of an issuer, the bondholder can negotiate with the issuer, participate in the restructure or bankruptcy proceeding.

Incorporate default risk into prices

Three main way to include the default risk in the bond pricing are through the pair intensity-recovery, the Z-spread or the historical default probabilities.

Intensity models a market implied probability. Z-spread measure a spread over the zero rates.

Market implied and real-world probabilities

Tho different approaches that can lead to different results.

Market implied probabilities

Inhomogeneous Poisson process with intensity lambda (≥0).
Survival probability P from t to T:

$$P(t,T) = \exp\left(-\int_t^T \lambda(s)ds\right)$$

Real-world probabilities

Time homogeneous Markov chain is used to model rating migrations and defaults.
The matrix P represent the 1-step (year) probabilities of each event.
In this case there are 3 possible states: investment grade, high yield, and default.
Default is an absorbing state.

$$P = \begin{array}{ccc} IG & HY & Def \\ \hline \pi_{1,1} & \pi_{1,2} & \pi_{1,3} \\ \pi_{2,1} & \pi_{2,2} & \pi_{2,3} \\ 0 & 0 & 1 \end{array}$$

Is P(t,T) decreasing in T?
Which property does each row of the P matrix satisfy?
How can one compute the matrix for the T-steps (years) probabilities?

Intensity-Recovery vs Z-Spread

Compare the two expressions. Which one do you expect to be used the most in real life?

Intensity-Recovery

$$\bar{C}(t) = \bar{c}\sum_{n=1}^{N}\delta_{n-1}\bar{B}(t,t_n) + \bar{B}(t,t_N) + \pi\sum_{n=1}^{N}e(0;t_{n-1},t_n) \qquad \bar{B}(t,T) = B(t,T)\exp\left(-\int_t^T \lambda(s)ds\right)$$

Z-Spread

$$\bar{C}(t) = \bar{c}\sum_{n=1}^{N}\delta_{n-1}\hat{B}(t,t_n) + \hat{B}(t,t_N) \qquad \hat{B}(t,T) = B(t,T)\exp\left(-\int_t^T z(s)ds\right)$$

Consider the case of constant intensity and constat Z-spread for the same bond. Recovery is given. Does a solution always exist? Is it unique?
Which one, between z-spread and intensity, do you expect to be the lowest value?

Bond price from transition matrix

It is an iterative procedure. We study zero coupon bonds.

1Y Bond IG

$$\bar{B}^{IG}(0,t_1) = (1-\pi_{1,3})B(0,t_1) + \pi_{1,3}B(0,t_1/2)\eta$$

2Y Bond IG

$$\bar{B}^{IG}(0,t_2) = \pi_{1,1}B(0,t_1)\bar{B}^{IG}(t_1,t_2) + \pi_{1,2}B(0,t_1)\bar{B}^{HY}(t_1,t_2) + \pi_{1,3}B(0,t_1/2)\eta$$

η is the recovery, which is supposed to be paid in the middle point of each period, in case of default.

Single factor model

How we can model market and idiosyncratic factor

Montecarlo simulations

For each obligor, at each time-step, one can simulate the firm value.
It is the sum of 2 components:

- the first is the market factor (y),
- the second is the idiosyncratic factor (ϵ).

ρ is the correlation with the market.

$$v_i = \sqrt{\rho} y + \sqrt{1-\rho}\epsilon_i$$

Thresholds for downgrade and default

One can define some thresholds such that, for each starting status i (n being default):

$P(v<K_{i,n}) = \pi_{i,n}$

$P(v<K_{i,j}) = \Sigma^{n}_{k=j}\pi_{i,k}$

$$P = \begin{array}{ccc} IG & HY & Def \\ \hline \pi_{1,1} & \pi_{1,2} & \pi_{1,3} \\ \pi_{2,1} & \pi_{2,2} & \pi_{2,3} \\ 0 & 0 & 1 \end{array}$$

We assume that all the variables simulated are independent identically distributed standard normal variables

03

Assignment Tasks

Six questions covering intensity, Z-spread, rating changes, and PnL VaR estimation

> *Use the data and bootstrapped curve from Assignment 0. Fill the gaps in the provided code, correct any errors, and comment your work in detail.*

Tasks 1 – 3 – Portfolio A (Beta)

Compute Z-spread and intensity under different assumptions

1

Z-spread - constant

Assume the Z-spread to be constant for each of the 2 bonds.
Compute the 2 Z-spreads.

2

Intensity - constant

Assume recovery rate 40%. Assume constant intensity for each of the 2 bond.
Compute the 2 intensities and the probability of default at 1 and 2 years.

3

Intensity - Piecewise constant

Assume recovery rate 40%. Assume piecewise constant intensity (constant over 1y period).
Compute the intensity implied in the 2 bonds prices and the probability of default at 1 and 2 years.

Tasks 4 – 5 – Portfolio A (Beta)

Credit shocks, conditional probabilities, market implied vs real world.

4

Creditworthiness deterioration scenario

Recompute the piecewise constant intensity and probabilities of default after the expectation of the medium-long term survival of Beta decreases. The 1y bond price is unchanged, but the 2y bond price drops to 97. Compare the result with the original scenario.

5

Compare with real world probability

Compute, from the matrix given in the code, the 1 and 2 years default probabilities.
Compute the default probability between the first and second year, conditional to the survival in the first year.
Compare with the respective quantities estimated in the creditworthiness deterioration scenario.

Tasks 6 – Portfolio B (100 issuers)

Simulate the portfolio with the single factor model

6 Single factor model simulations

Suppose all the issuers of the bonds in the portfolio are IG.
Consider the real-world probabilities matrix and recovery rate 40%.

a) Evaluate the portfolio MtM
b) Derive the 1y forward price of the bond, in case the issuer (i) is still IG, (ii) has been downgraded to HY, (iii) defaulted.
c) Execute Monte-Carlo simulations with (i) $\rho = 0\%$, (ii) $\rho = 25\%$ (iii) $\rho = 50\%$ to evaluate the value of each firm in 1 year. Use 1 000 000 scenarios. Count the average number of downgrades and defaults and estimate the 99% VaR of your PnL considering (i) only defaults losses or (ii) both default losses and PnL from migrations.
d) Compare the different VaR computed, study the effect of migration risk and correlation impact.

Deliverables & Practical Notes

1. Deliver by Friday 27 March, 18:00 to financial.engineering.polimi@gmail.com with subject "RM: Assignment 2, Group #".
2. Submit a short PDF with results (Z-spread, Intensities, PD, MtM, VaR) and list any errors found in the code.
3. Submit the Python code (any IDE). Comment in detail and use descriptive variable names.
4. Use the bootstrapped curve from Assignment 0. In case of inconsistencies between the PDF instructions and the code, follow the PDF.
5. Ignore the coupon accrual and their recovery. For the intensity cases, assume recovery paid at the end of each period. For the real-world cases, assume the recovery paid in the middle of each period.